Prediction of Mental Workload in Single and Multiple Tasks Environments

Cognitive Load

Prédire la charge mental en prenant en considération des caractéristiques individuel
Modélisation de la charge mental en 2 parties :

- Effective workload et ineffective workload

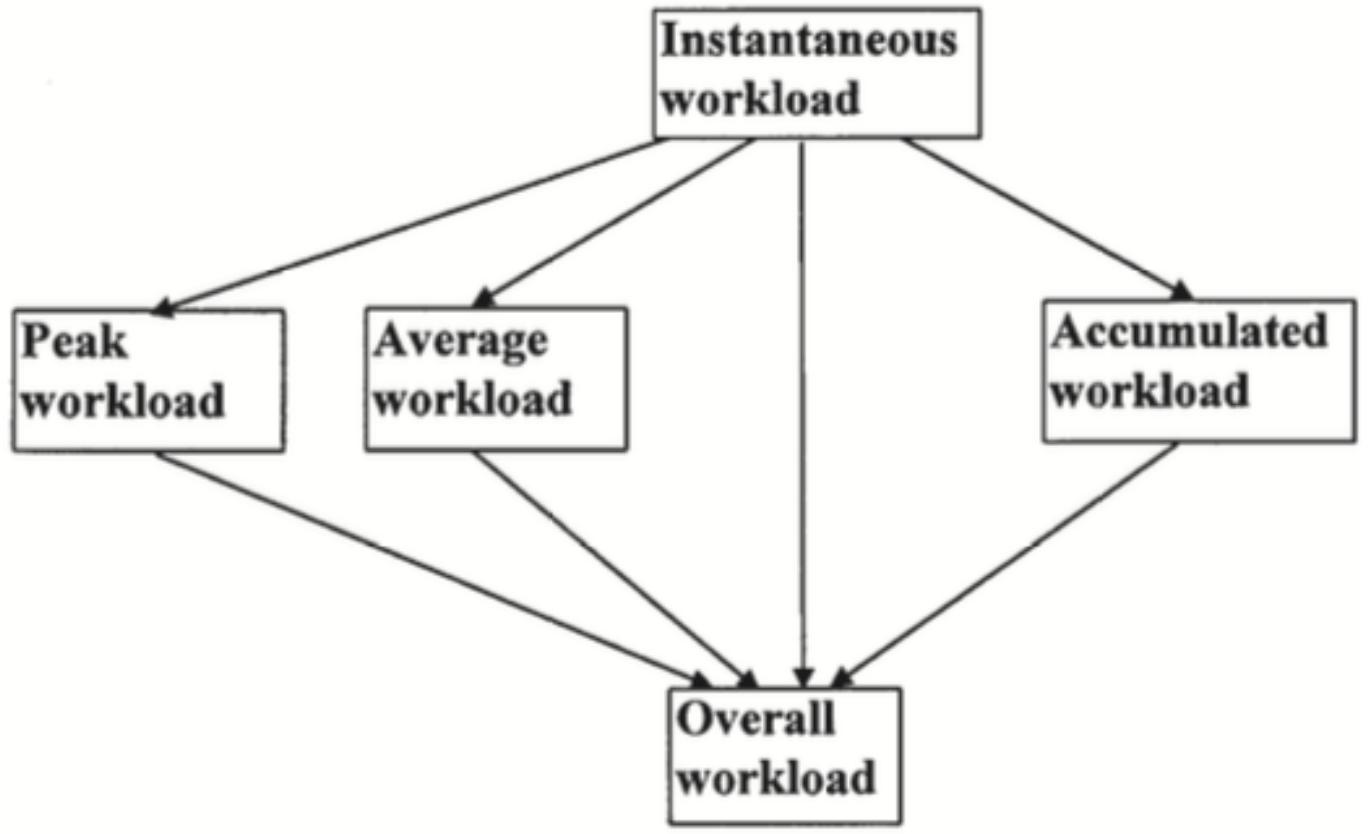

FIGURE 1 Conceptual framework of mental workload types.

The relationship between the various workload measures is mathematically defined here:

$$W_p = Max\{W_1(t)\};$$

$$W_2(t) = \int_0^t W_1(u)du;$$

$$W_3(t) = \frac{1}{t}\cdot W_2(t);$$

$$W_o(t) = F_1[W_1(t)] = F_2[W_2(t), W_3(t)]$$

Cumulative workload: ils considèrent la cumulation du workload à chaque moment de la tache au sens de la somme pour défnir le **workload de la tache**

Dans notre problème le workload de la tache est **fixé** par l'instance :
- Au lieu de faire la **somme** sur chaque instant d'éxécution de la tache pour avoir le **workload de la tache**
- On fait la somme sur les workload de chasue tache fait par l'opérateur pour avoir son **workload cumulé de la periode**

hypothèse simplificatrice pour le moment

H1

Workload is necessarily task and person specific

- effective workload : workload que génere la tache naturellement -> **task load**
- Ineffective workload : tend a se réduire avec l'apprentissage et répétition de la tache -> **pour notre problème temps de la tache deterministe**

Tc: Difficulty of task \in {1, 2, 3, 4} **Pk: Level of Worker \in [1, 4]**

In our study, task complexity (Tc) and task type (Tt), which are task related factors, were selected to estimate the effective workload. The personal knowledge (Pk) and physical skill (Pp), which are individual related factors, were selected to estimate the ineffective workload.

If we further assume the linear correlation exists between the workload and the selected, then

$$W = W_{eff} + W_{ineff} = a*Tc + b*Tt + c*Pk + d*Pp + f$$

where a, b, c, d, and f are constants, and the parameters are defined as before.

Task complexity T_c was obtained using Zhao's (1992) approach. Task types were categorized as single task, system-paced multitask, and self-paced multitask. This classification aims to study the mental workload correlation under both single- and multitask environments. Domain knowledge of individuals (P_k), such as knowledge of HTML, were obtained using a designed questionnaire. Physical skill (P_p), such as typing and mouse movement ability, was obtained from a skill test.

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Cognitive Load

$D_{task} \in \{1,2,3,4\}$ Difficulty of the Task

$L_i \in [1;4]$ Level/skills of the worker

$a, b_i \in \{0,1\}$ Coeficient of Each part

W = W_eff + W_ineff

Alone ⟶ task — *No mental Workload*

Teaching ⟶ task

$a \cdot D_{task} + b_{apprenti} \cdot (5 - L_{apprenti})$

$a \cdot D_{task} + b_{tutor} \cdot (5 - L_{tutor})$

Collaboration of k_1 and k_2 ⟶ task

$a \cdot D_{task} + b_{collab} \cdot (5 - L_{k_i})$

Par **H1** pour savoir la charge mental de chaque worker à la fin de la période on agrègent ce que chaque tache lui a engendré avec une somme